Parnassus: Neural Fast Simulation and Reconstruction for High Energy Physics

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Abstract

We present the public software release of Parnassus, a pure-Python, GPU-compatible framework for neural fast detector simulation and reconstruction in High Energy Physics. Parnassus replaces the computationally expensive Geant4 simulation and reconstruction chain with conditional flow matching models that map truth-level particles directly to detector-level particle-flow objects. The framework is both process-agnostic and detector-agnostic: users select a detector model—analogous to choosing a detector card in Delphes—and the same pre-trained model can be applied to arbitrary physics processes without retraining. Unlike C++/ROOT-based tools such as Delphes, Parnassus runs natively on GPU (CUDA and Apple MPS) and requires no ROOT installation. The current release ships with a CMS detector model trained on 2011 open data; additional detector models will be provided in future releases. We describe the installation, command-line and Python API, configuration system, and demonstrate the framework on Standard Model and BSM processes.

Keywords: fast simulation, detector simulation, particle flow, generative models, flow matching, diffusion models, high energy physics

PROGRAM SUMMARY

Program title:	Parnassus
CPC Library link to program files:	(to be added by Technical Editor)
Developer's repository link:	https://github.com/parnassus-hep/parnassus
Licensing provisions:	MIT
Programming language:	Python ($\geq$ 3.12)
Nature of problem:	Full detector simulation and reconstruction based on Geant4 is computationally expensive, often dominating the computing budget of large-scale High Energy Physics analyses. A fast, accurate surrogate is needed that maps truth-level (generator-level) particles directly to detector-level particle-flow objects, preserving the fidelity of the full chain while being orders of magnitude faster.
Solution method:	Parnassus trains conditional flow matching models [7, 8] on full simulation and reconstruction data. Three cascaded diffusion models generate (1) event-level quantities, (2) per-particle kinematics and classification, and (3) track impact parameters, conditioned on truth-level particle properties. Generation is followed by physics-based post-processing (jet clustering via FastJet [17], lepton isolation). Users select a detector model—analogous to a Delphes card—to target a specific experiment.

Additional comments including restrictions and unusual features:

The framework is implemented entirely in Python with PyTorch and runs natively on CPU, CUDA GPU, and Apple MPS. It has no dependency on ROOT; output to ROOT format is optionally handled via `uproot` [24]. The current release ships with a CMS detector model trained on 2011 open data [6]; additional detector models for other experiments will be added in future releases. The CMS model supports up to 400 truth particles per event; events exceeding this limit retain the 400 highest-p_{T} particles.

1 Introduction

Parnassus is a **pure-Python**, **fully GPU-compatible** framework for **fast detector simulation and reconstruction** in High Energy Physics (HEP). It replaces the computationally expensive full GEANT4 [3] simulation and reconstruction chain with neural network-based generative models that learn the mapping from truth-level (generator-level) particles directly to detector-level particle-flow [4, 5] (pflow) objects, bypassing intermediate steps such as hit simulation, digitisation, and track reconstruction. The framework is both **process-agnostic** and **detector-agnostic**: the same architecture supports multiple detector models, and for each detector the pre-trained model can be applied to arbitrary physics processes by simply providing different input truth events.

Unlike traditional fast simulation tools such as DELPHES [25], which are written in C++, require the ROOT framework [19], and run only on CPU, Parnassus is implemented entirely in Python with PyTorch [22] and runs natively on GPU (CUDA and Apple MPS), enabling seamless integration into modern Python-based analysis workflows and significant acceleration on GPU hardware. ROOT I/O is supported via `uproot` [24] for writing output files, but is entirely optional — Parnassus has no dependency on ROOT itself.

Users select a **detector model** — analogous to choosing a detector card in DELPHES — to target a specific experiment. The current release ships with a CMS detector model trained on 2011 open data [6]; additional detector models for other experiments will be added in future releases.

The physics methodology underlying Parnassus has been described in detail in Refs. [1, 2], where we demonstrated that the approach reproduces full CMS simulation and reconstruction across a wide range of Standard Model and BSM processes. In this document we focus on the **public software release** — its installation, usage, and API — to enable the broader HEP community to adopt neural fast simulation and reconstruction in their workflows.

The neural backbone uses conditional flow matching [7, 8] to train continuous-time diffusion models [9, 10] that are sampled via Euler integration at inference time. The models are conditioned on truth-level particle properties, making the generation inherently process-agnostic without requiring retraining for new physics scenarios.

1.1 Design goals

Speed Orders of magnitude faster than full simulation and reconstruction by replacing the detector response with neural diffusion models. Fully GPU-compatible for additional acceleration.

Pure Python Implemented entirely in Python with no compiled dependencies beyond PyTorch. Unlike C++/ROOT-based tools such as DELPHES [25], Parnassus integrates naturally into modern Python analysis ecosystems. ROOT output is available via `uproot` but is not required.

Fidelity Detector models are trained on full simulation and reconstruction data to reproduce realistic detector-level distributions including particle kinematics, vertex positions, impact parameters, and particle identification. The current CMS detector model, trained on 2011 open data [6], shows excellent agreement with full CMS simulation across diverse processes [1, 2].

Flexibility Accept truth-level input from any source — PYTHIA8 [11] on-the-fly generation, HepMC [12] files from external generators (MADGRAPH5_aMC@NLO [13], SHERPA [14], HERWIG [15], POWHEG [16]), or custom formats. Users select a detector model to target a specific experiment, analogous to choosing a detector card in DELPHES.

Completeness Full post-processing pipeline including jet clustering (FASTJET [17]) and lepton isolation, with optional output to ROOT format via `uproot` [24].

1.2 Output quantities

Given a set of truth-level particles for each event, Parnassus generates the quantities listed in Table 1.

Table 1: Output quantities produced by Parnassus for each event.

Output	Description
Particle-flow particles	Full kinematics (p_T, η, ϕ), vertex (v_x, v_y, v_z), class, PDG ID
Impact parameters	d_0, z_0 and their errors for charged particles
Jets	Clustered with configurable algorithms (anti-k_t [18], gen-k_t); substructure (D_2 [20], C
Lepton isolation	Isolation scores and p_T sums in configurable ΔR cones
Event-level quantities	H_T, $\vec{E}_\mathrm{T}^\mathrm{miss}$ components, particle multiplicity

2 Architecture overview

Parnassus uses a three-stage neural generation pipeline followed by physics-based post-processing. The architecture is illustrated in Fig. 1.

All three neural models are **diffusion models** trained with conditional flow matching [7, 8] and sampled via Euler integration. Each model is conditioned on truth-level particle information, making the generation process inherently dependent on the input physics process without being tied to any specific one.

2.1 Neural model details

Table 2 summarises the three neural models and their input/output specifications.

The number of diffusion steps is configurable at runtime via the `--num_steps` (`-n`) flag, allowing users to trade generation speed for quality.

3 Installation and quick start

3.1 Installation

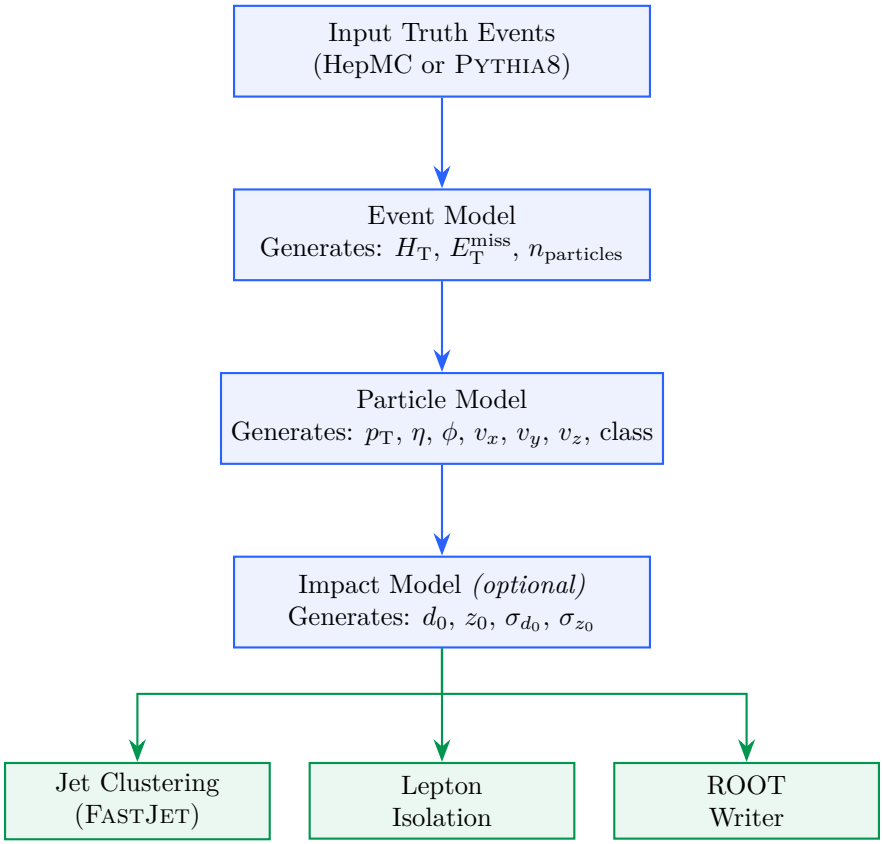

Figure 1: Parnassus generation pipeline. Three neural diffusion models (blue) produce detector-level quantities from truth-level input. Physics-based post-processing stages (green) compute derived quantities and write the final output.

Table 2: Neural model specifications (shown for the CMS 2011 detector model). The same architecture applies to all detector models.

Model	**Input Context**	**Output Variables**	**Sampler**
Event	Truth particle kinematics (p_T^{rel}, η, ϕ, v_x, v_y, v_z, class), global sums (H_T, MET, n_{truth})	pflow H_T, MET_x, MET_y, n_{pflow}	Euler, 50 steps
Particle	Same + event model outputs	pflow p_T^{rel}, η, ϕ, v_x, v_y, v_z, class	Euler (reverse time), 50 steps
Impact	Same + event model outputs	d_0, z_0, σ_{d_0}, σ_{z_0}	Euler, 50 steps

```bash
# Clone the repository
git clone https://github.com/parnassus-hep/parnassus.git
cd parnassus

# Install with pip (requires Python 3.12)
pip install .

# For development
pip install -e ".[dev]"
```

Listing 1: Installation commands.

3.2 Requirements

- Python $\geq 3.12, < 3.13$
- PyTorch [22] $\geq 2.6.0$
- Pythia8mc [11] 8.316.0
- PyHepMC [12] $\geq 2.14.0$
- FastJet [17] $\geq 3.4.3.1$
- EnergyFlow [23] $\geq 1.4.0$
- Uproot [24] $\geq 5.5.2$ *(optional, for ROOT output)*

3.3 Initialise a configuration file

```bash
parnassus init .
```

This copies the default `default_config.yaml` to the current directory. The default configuration uses the `cms_2011_flow_v00` detector model with anti-k_t jet clustering and lepton isolation. To target a different detector, change the `generator.name` field to the desired detector model.

3.4 Run with a HepMC input file

```bash
parnassus run \
  -c default_config.yaml \
  -i events.hepmc \
  -ne 1000 \
  -bs 2000 \
  -o output.root
```

Listing 2: Basic command-line invocation.

The available command-line flags are listed in Table 3.

3.5 Run with Pythia8 on-the-fly generation

To generate truth-level events on-the-fly with Pythia8, create a `.cmnd` steering card and pass it as the input file. Parnassus detects `.cmnd` files automatically:

Table 3: Command-line flags for `parnassus run`.

Flag	Long form	Description
`-c`	`--config`	Path to YAML configuration file
`-i`	`--input_path`	Input HepMC or Pythia `.cmnd` file
`-o`	`--output_path`	Output ROOT file path
`-ne`	`--num_events`	Number of events to process
`-bs`	`--batch_size`	Batch size for neural generation
`-n`	`--num_steps`	Diffusion sampler steps (default: 50)

```
parnassus run \
  -c default_config.yaml \
  -i ttbar.cmnd \
  -ne 5000 \
  -bs 2000 \
  -o ttbar_fastsim.root
```

4 Demonstration: SM and BSM processes

The neural models are **process-agnostic** — they learn the detector response as a function of truth-level particle properties, not the physics process itself. We have previously demonstrated [1, 2] that Parnassus reproduces full simulation and reconstruction across a wide range of SM and BSM processes. Here we show two representative examples using the CMS detector model to illustrate the API in action: a Standard Model process ($H \to ZZ \to 4\ell$) and a BSM exotic Higgs decay ($H \to aa \to gg\mu\mu$). The same workflow applies to any detector model by changing the `generator.name` in the configuration.

The current CMS detector model supports up to 400 truth particles per event. For events exceeding this limit, Parnassus retains the 400 highest-p_{T} particles so that no events are discarded.

4.1 Standard Model: $H \to ZZ \to 4\ell$

We process 100 events from a HepMC input file produced by Pythia8 [11]:

```
parnassus run \
  -c default_config.yaml \
  -i h4l_events.hepmc \
  -ne 100 \
  -o h4l_output.root
```

4.2 BSM: $H \to aa \to gg\mu\mu$

For the BSM example, we generate exotic Higgs decays on-the-fly with Pythia8. The Higgs decays to a pair of light pseudoscalars a ($m_a = 20\,\mathrm{GeV}$), each decaying to either gg or $\mu^+\mu^-$, producing a mixed final state of jets and muon pairs:

```
! haa_ggmumu.cmnd -- BSM exotic Higgs decay
Beams:eCM = 13000.
Beams:idA = 2212
Beams:idB = 2212
```

```
! BSM Higgs production via gluon fusion
Higgs:useBSM = on
HiggsBSM:gg2H2 = on
35:m0 = 125.0                          ! Higgs mass

! SM-like couplings for production
HiggsH2:coup2u = 1.0
HiggsH2:coup2d = 1.0
HiggsH2:coup2A3A3 = 1.0                ! H -> aa coupling

! Force H -> a a only
35:onMode = off
35:onIfAll = 36 36

! Light pseudoscalar a properties (PDG ID 36)
36:m0 = 20.0                           ! mass = 20 GeV
HiggsA3:coup2d = 1.0
HiggsA3:coup2u = 1.0
HiggsA3:coup2l = 1.0

! Force a -> gg or mumu only
36:onMode = off
36:onIfAny = 21 13
```

Listing 3: Pythia8 steering card for BSM $H \to aa \to gg\mu\mu$.

```
parnassus run \
  -c default_config.yaml \
  -i haa_ggmumu.cmnd \
  -ne 1000 \
  -o haa_ggmumu_output.root
```

This demonstrates running Parnassus on a BSM signal process not present in the training data. The mixed $gg\mu\mu$ final state probes the model's ability to simultaneously generate hadronic jets and isolated muons within the same event.

4.3 Kinematic distributions

Fig. 2 compares the particle-flow kinematic distributions for both processes. The p_{T} spectra follow the expected steeply falling shape, the η distributions reflect the detector acceptance, and the multiplicity distributions reflect the different underlying event structures. The BSM $H \to aa$ process shows higher multiplicity due to the hadronic activity from the $a \to gg$ decays.

4.4 Event display

Fig. 3 shows a representative $H \to 4\ell$ event in the η–ϕ plane, illustrating how Parnassus transforms truth-level particles into detector-level pflow objects.

4.5 Aggregate statistics

Table 4 compares the aggregate statistics for both processes. The BSM process shows higher truth-particle multiplicity (163 ± 70) due to the $a \to gg$ hadronic decays, while the pflow output multiplicity is comparable across both processes.

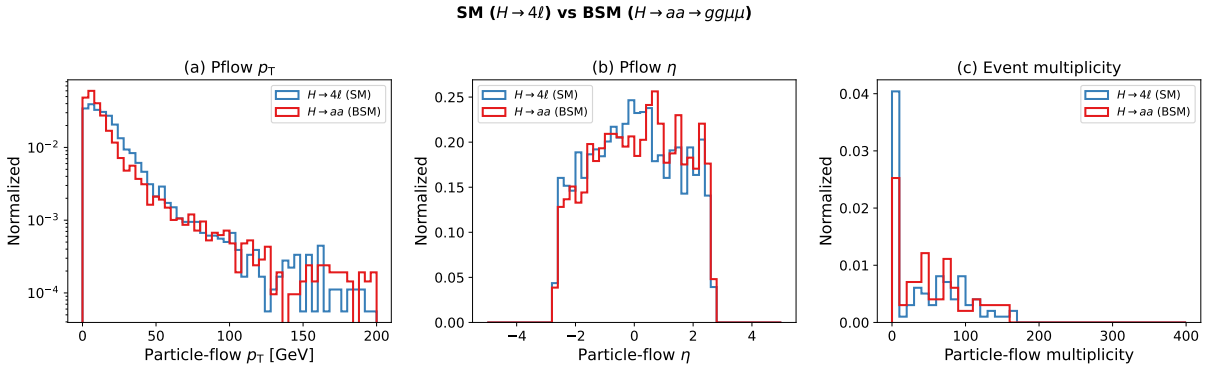

Figure 2: Particle-flow kinematic distributions comparing SM ($H\to4\ell$, 99 events) and BSM ($H\to aa\to gg\mu\mu$, 99 events) processes. (a) p_{T} in log scale, (b) pseudorapidity η, (c) particle-flow multiplicity per event. All distributions are normalised to unit area.

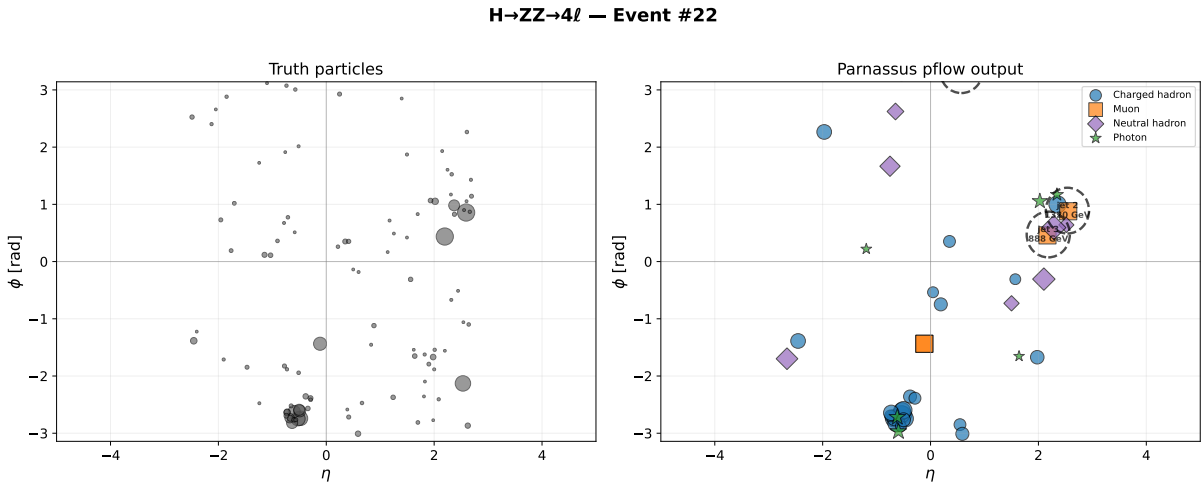

Figure 3: Event display for $H \to ZZ \to 4\ell$: truth particles (left) and Parnassus pflow output (right) in the η–ϕ plane. Marker size scales with p_{T}; dashed circles indicate anti-k_t jets ($R = 0.5$). Particle classes: charged hadrons (blue circles), electrons (red triangles), muons (orange squares), neutral hadrons (purple diamonds), photons (green stars).

Table 4: Aggregate statistics for SM and BSM processes (mean ± std).

Quantity	$H\to4\ell$ **(99 evt)**	$H\to aa$ **(99 evt)**
Truth particles/event	97.2 ± 46.0	163.4 ± 69.9
Pflow particles/event	46.3 ± 47.2	53.6 ± 46.4

4.6 Particle class distribution

Fig. 4 compares the particle class composition between the two processes. Both show similar class fractions dominated by charged hadrons (∼60%), consistent with the process-agnostic nature of the model.

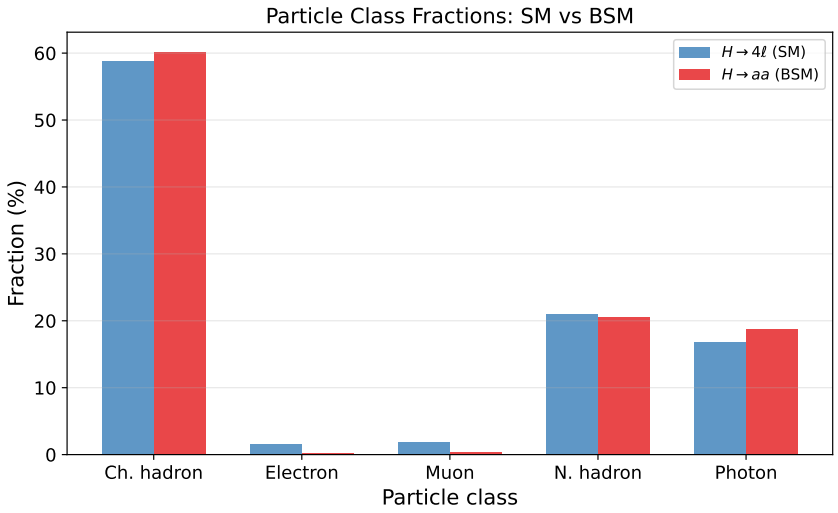

Figure 4: Particle class fractions for SM ($H \to 4\ell$) and BSM ($H \to aa \to gg\mu\mu$) processes. The model produces consistent class compositions regardless of the input physics process.

5 Diffusion steps convergence

The number of diffusion steps N controls the trade-off between generation quality and speed. Fig. 5 shows the convergence of particle-flow kinematic distributions as a function of N for the $H \to 4\ell$ sample.

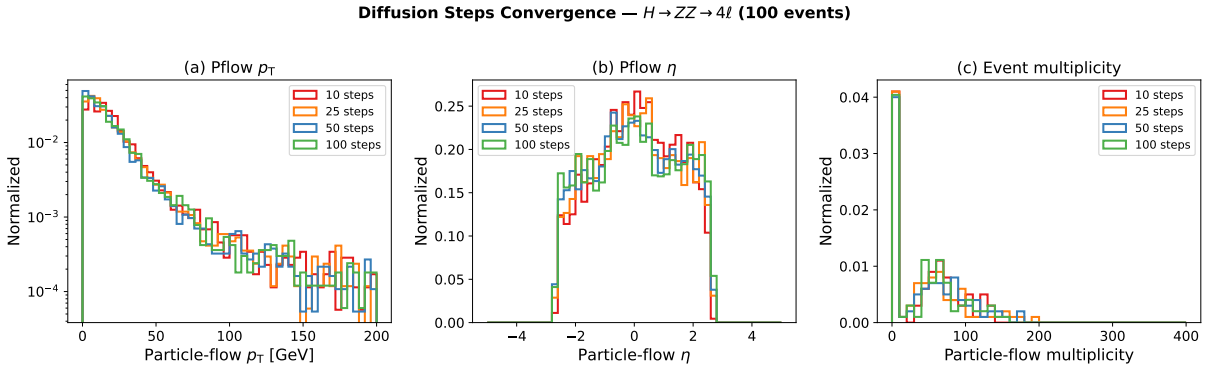

Figure 5: Diffusion steps convergence for $H \to ZZ \to 4\ell$ (100 events). (a) Pflow p_{T}, (b) pflow η, (c) event multiplicity. Distributions are stable from $N = 25$ onward.

Fig. 6 shows that particle class fractions are also stable across step counts.

Table 5 summarises the generation time as a function of diffusion steps on CPU. The default of 50 steps provides a good balance of quality and speed. For quick validation runs, 25 steps yields nearly identical distributions at lower cost.

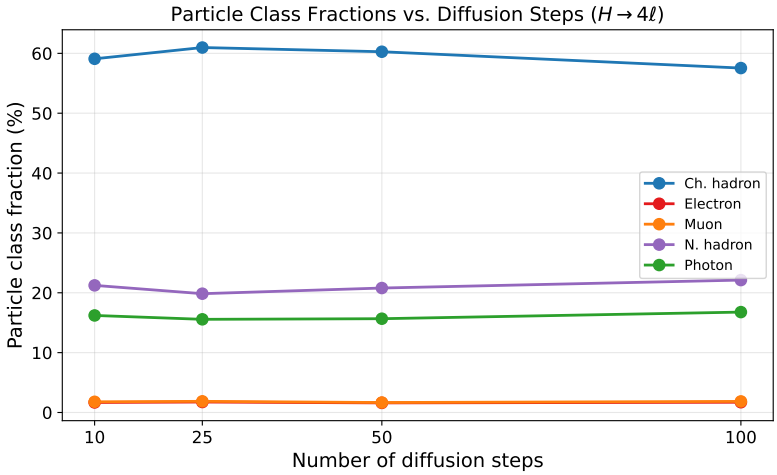

Figure 6: Particle class fractions vs. number of diffusion steps. All classes converge by $N = 25$.

Table 5: Generation time vs. diffusion steps (100 events, CPU).

Steps	Wall time
10	∼1:30
25	∼1:30
50	∼2:10
100	∼5:20

6 Python API

For programmatic usage, Parnassus provides a clean Python API:

```python
from pathlib import Path
from parnassus.configs import Config
from parnassus.pipelines import (
    GenerationPipeline,
    JetClusteringPipeline,
    IsolationPipeline,
)
from parnassus.configs.pipeline import (
    JetClusteringConfig,
    IsolationConfig,
)
from parnassus.writers import RootWriter

# Load configuration
config = Config.from_yaml("my_config.yaml")

# Override settings programmatically
config.dataset_config.file_path = (
    Path("my_events.hepmc").absolute()
)
config.dataset_config.num_events = 5000

# Run generation
pipeline = GenerationPipeline(config)
gen_events, accessors_dict = pipeline.run()
```

```python
# Post-processing
for pipeline_config in config.pipeline_configs:
    if isinstance(pipeline_config, JetClusteringConfig):
        jet_pipeline = JetClusteringPipeline(pipeline_config)
        jet_pipeline.process(gen_events)
    elif isinstance(pipeline_config, IsolationConfig):
        iso_pipeline = IsolationPipeline(pipeline_config)
        iso_pipeline.process(gen_events)

# Inspect generated data
for event in gen_events[:5]:
    print(f"Event {event.event_number}:")
    n_truth = len(event.truth_particles.pt)
    n_pflow = len(event.pflow_particles.pt)
    print(f"  Truth particles: {n_truth}")
    print(f"  Pflow particles: {n_pflow}")
    print(f"  HT = {event.ht:.1f} GeV")

# Write to ROOT
writer = RootWriter(config.writer_config)
writer.write(gen_events)
```

Listing 4: Programmatic usage of the generation pipeline.

6.1 Reading output files

```python
import uproot
import numpy as np

f = uproot.open("output.root")
tree = f["Parnassus"]

# Access pflow jet pT
jet_pt = tree["PflowJetsAntiKt05.pt"].array()

# Access electron isolation
e_iso = tree["Electrons.iso_var"].array()

# Access impact parameters
d0 = tree["Pflow.d0"].array()
z0 = tree["Pflow.z0"].array()
```

Listing 5: Reading the output ROOT file with `uproot`.

7 Configuration reference

```yaml
# -- Dataset ---------------------------------------------
dataset:
  file_path: ""               # Path to .hepmc or .cmnd file
  num_events: 1000            # Number of events to process
  entry_start: 0              # Starting event index (HepMC)

# -- Generator -------------------------------------------
generator:
```

```yaml
  type: "neural"              # "neural" or "parametric"
  name: "cms_2011_flow_v00"   # Detector model (like a Delphes card)
  num_steps: 50               # Diffusion steps (neural only)
  batch_size: 2000            # Events per batch
  device: "cpu"               # "cpu", "cuda", or "mps"

# -- Post-processing Pipelines ------------------------
pipelines:
  TruthJetsAntiKt05:
    type: "cluster"
    collection: truth
    dr: 0.5
    algorithm: antikt
    pt_min: 10
    nconst_min: 2
  PflowJetsAntiKt05:
    type: "cluster"
    collection: pflow
    dr: 0.5
    algorithm: antikt
    pt_min: 10
    nconst_min: 2
  ElectronIsolation:
    type: "isolation"
    collection: "electrons"
    dr: 0.4
  MuonIsolation:
    type: "isolation"
    collection: "muons"
    dr: 0.4

# -- Output -------------------------------------------
output:
  file_path: ""               # Output ROOT file path
  format: default
```

Listing 6: Annotated default configuration.

8 Conclusion

We have presented the public release of Parnassus, a pure-Python, GPU-compatible framework for neural fast simulation and reconstruction in High Energy Physics. Unlike traditional C++/ROOT-based tools such as Delphes [25], Parnassus runs entirely in Python with PyTorch, supports GPU acceleration (CUDA and Apple MPS), and requires no ROOT installation — ROOT output is optionally handled via `uproot`. The software provides a simple command-line and Python API for generating detector-level particle-flow objects from truth-level input, with full post-processing including jet clustering and lepton isolation.

The framework is both process-agnostic and detector-agnostic. Users select a detector model — analogous to choosing a detector card in Delphes — to target a specific experiment. The current release ships with a CMS detector model validated against full CMS simulation across diverse SM and BSM processes [1, 2]; additional detector models will be provided in future releases.

The source code is publicly available at https://github.com/parnassus-hep/parnassus under the MIT license.

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